

Applications of special relativity

What gives gold its colour?

In discussions of special relativity, you occasionally encounter a claim like, “The effects of special relativity only matter to particle physicists and others working with extreme energies and velocities. Relativity has no consequences in everyday life.” Well, these days, anybody who relies upon the Global Positioning System (GPS) to navigate their car or the airliner in which they’re travelling uses both special and general relativity, because without correction for their effects, GPS would be so inaccurate as to be useless. But GPS is a recent innovation, and the relativistic corrections are both complicated and hidden from the user in the software in the receiver and on board the satellites. But there’s an effect of special relativity which was observed, if not understood, by the ancients: the yellow gleam of gold.

With an atomic number of 79, gold is in the last row of the periodic table containing stable elements, and only four stable elements (mercury, thallium, lead, and bismuth) have greater atomic number. With 79 protons in its nucleus, the electrons of the gold atom are subjected to an intense electrostatic attraction. Using the naïve Bohr “solar system” model of the atom for the moment, electrons in the 1s orbital, closest to the nucleus, would have to orbit with a velocity v of $1.6 \times 10^8 \text{ m s}^{-1}$ to have sufficient kinetic energy to avoid “falling into” the nucleus. This is more than half the speed of light. At these speeds the momentum of the electron increases causing a relativistic contraction of its orbit.

The colour of metals such as silver and gold is mainly due to absorption of light when a d electron jumps to an s orbital. For silver, the $4d \rightarrow 5s$ transition has an energy corresponding to ultraviolet light, so frequencies in the visible band are not absorbed. With all visible frequencies reflected equally, silver has no colour of its own; it’s silvery. In gold, however, relativistic contraction of the s orbitals causes their energy levels to shift closer to those of the d orbitals (which are less affected by relativity). This, in turn, shifts the light absorption from the ultraviolet down into blue visual range. A substance that absorbs blue light will reflect the rest of the spectrum: the reds and greens which, combined, result in the yellowish hue we call golden.

Special relativity is also responsible for gold’s resistance to tarnishing and other chemical reactions. Chemistry is mostly concerned with the electrons in the outermost orbitals. With a single 6s electron, you might expect gold to be highly reactive; after all, caesium has the same 6s outer shell, and it is the most alkaline of natural elements: it explodes if dropped in water, and even reacts with ice. Gold’s 6s orbital, however, is relativistically contracted toward the nucleus, and its electron has a high probability to be among the electrons of the filled inner shells. This, along with the stronger electrostatic attraction of the 79 protons in the nucleus, reduce the “atomic radius” of gold to about half of that for caesium with its 55 protons and electrons. The gold atom is almost 50% heavier, yet only a little over half the size of caesium giving gold its high density. Only the most reactive substances can tug gold’s 6s¹ electron out from where it’s hiding among the others. The colour of gold and its immunity to tarnishing and corrosion are consequences of special relativity.



GPS systems

The Global Positioning System (GPS) is based on an array of 24 satellites, each carrying a precise atomic clock, orbiting the Earth. Using a hand-held GPS receiver which detects radio emissions from any of the satellites which happen to be overhead, users of even moderately priced devices can determine latitude, longitude and altitude to an accuracy currently reaching 15 metres, and local time to 50 billionths of a second. GPS has applications in aircraft navigation, oil exploration, wilderness recreation, bridge construction, sailing, and interstate trucking.

But in a relativistic world, things are not simple. The satellite clocks are moving at $14,000 \text{ km h}^{-1}$ in orbits that circle the Earth twice per day, much faster than clocks on the surface of the Earth, and Einstein's theory of special relativity says that these rapidly moving clocks tick more slowly, by about seven microseconds (millionths of a second) per day.

Also, the orbiting clocks are 20,000 km above the Earth, and experience gravity that is one-quarter that on the ground. Einstein's general relativity theory says that gravity curves space and time, resulting in a tendency for the orbiting clocks to tick slightly faster, by about 45 microseconds per day. The net result is that time on a GPS satellite clock advances faster than a clock on the ground by about 38 microseconds per day.

To determine its location, the GPS receiver uses the time at which each signal from a satellite was emitted, as determined by the on-board atomic clock and encoded into the signal, together with the speed of light, to calculate the distance between itself and the satellites it communicated with. The orbit of each satellite is known accurately. Given enough satellites, it is a simple problem in Euclidean geometry to compute the receiver's precise location, both in space and time. To achieve a navigation accuracy of 15 metres, time throughout the GPS system must be known to an accuracy of 50 nanoseconds, which simply corresponds to the time required for light to travel 15 metres.

But at 38 microseconds per day, the relativistic offset in the rates of the satellite clocks is so large that, if left uncompensated, it would cause navigational errors that accumulate faster than 10 km per day! GPS accounts for relativity by electronically adjusting the rates of the satellite clocks, and by building mathematical corrections into the computer chips which solve for the user's location. Without the proper application of relativity, GPS would fail in its navigational functions within about 2 minutes.

Experimental evidence for time dilation

One of the first experiments to provide evidence of time dilation was done over seventy years ago by detecting muons. Muons are particles created when incoming cosmic rays collide with air molecules in the outer reaches of the atmosphere. A constant stream of these particles travels towards the surface of the Earth at speeds very close to the speed of light. They are unstable particles, with a "half-life" of about 1.5 microseconds, which means that if you start with a thousand, 1.5 microseconds later you will have about 500 and so on.

In 1941, a detector placed near the top of Mount Washington (at 2000 m above sea level) measured muon flux of about 570 per hour. It would be expected that the number of muons would decrease as they fall, so if we move the detector to a lower altitude we expect it to detect fewer muons.

Knowing the half-life, and given that 570 per hour hit a detector near the top of Mount Washington, it would be expected that about 35 muons per hour would survive down to sea level. But with the detector at sea level about 400 muons per hour were detected! How do we explain the difference? The reason they didn't decay is that in their frame of reference, much less time had passed. The actual speed of the muons is about $0.994c$, which corresponds to a time dilation factor of about 9. This means that in the time it takes to travel from the top of Mount Washington to sea level (6 microseconds), the time registered on the muons internal clock is about 0.67 microseconds ($6/9$) microseconds. In this time, only about one-quarter of them would decay leaving $\frac{3}{4}$ or about 430 to reach the detector.

Length contraction also plays its part. From the muon's point of view Mount Washington is less than 240m high. This explains why they can cover the distance so quickly.

In 1979 scientists carried out an experiment using a particle accelerator at CERN. They reported a similar experiment with muons accelerated to speeds $0.9994c$. Trapped in a particle accelerator, muons were observed in the lab to have 29.3 times their rest half-life, completely consistent with time dilation.

Chapter 14: Special relativity

Special relativity explained

Notes

Main points

- Observations of objects travelling at very high speeds cannot be explained by Newtonian physics. These include the dilated half-life of high-speed muons created in the upper atmosphere, and the momentum of high-speed particles in particle accelerators.
- Einstein's special theory of relativity predicts significantly different results to those of Newtonian physics for velocities approaching the speed of light.
- The special theory of relativity is based on two postulates: that the speed of light in a vacuum is an absolute constant, and that all inertial reference frames are equivalent.
- Motion can only be measured relative to an observer; length and time are relative quantities that depend on the observer's frame of reference.

Equations:

The frequency f , wavelength λ , and wave speed v are related by the equation: $v = f\lambda$

The relationship between mass and energy is: $E = mc^2$

In a nuclear reaction: $E = \Delta m c^2$

The factor v/c is often referred to as β where: $\beta = \frac{v}{c}$

The Lorentz factor: $\gamma = \frac{1}{\sqrt{1 - \beta^2}}$

If L_0 is the length of an object measured by an observer in the object's reference frame and L is the length as measured by the observer in a stationary reference frame then the two measurements are related as follows: $L = L_0 / \gamma$

$t = \gamma t_0$

Here t_0 is the time as measured in the frame of the moving 'light-clock' and t is the time as measured by the stationary observer.

Relative velocities: $u' = \frac{u - v}{1 - \frac{uv}{c^2}}$ $u = \frac{u' + v}{1 - \frac{u'v}{c^2}}$

Relativistic energy and momentum: $E_k = \gamma mc^2$ $p = \gamma mv$

To find a photon's momentum p , we can combine Planck's formula ($E = hf$) and Einstein's mass/energy formula ($E = mc^2$):

$$p = mc = \frac{E}{c} = \frac{hf}{c} = \frac{h}{\lambda}$$

Special relativity

For a long time many scientists assumed that there must be some kind of substance through which light propagated similar to that observed with waves in water and with sound waves. They called this unknown substance the 'ether'. In 1887 Albert Michelson and Edward Morley carried out an experiment in which they tried to show the motion of Earth relative to the ether by measuring changes in light speed in different directions. To their surprise they found no change in the light speed regardless of the relative motion between Earth and the source of light or the ether.

Their experiment was a 'failure' but this 'failure' enabled scientists to abandon their ideas of the theory of 'ether' and to think differently. Without the ether theory there was now no absolute reference frame to determine what is at rest and what is moving. At this point Einstein started his work on special relativity.

Albert Einstein

Albert Einstein was born in Ulm, Germany on March 14, 1879. As a child, Einstein developed a curiosity for understanding the mysteries of science. Moving to Italy and then to Switzerland, he graduated from high school in 1896. In his later years, Einstein would write about two events that had a marked effect on his childhood. One was an encounter with a compass at age five, where he marvelled at the invisible forces that turned the needle. The other was at age 12, when he discovered a book of geometry which he read over and over.

In 1905, while working as a patent clerk in Bern, Switzerland, Einstein had what came to be known as his “Annus Mirabilis” or “miracle year”. It was during this time that he obtained his doctorate degree and published four of his most influential research papers, including the special theory of relativity. This theory and his now world famous equation “ $E = mc^2$ ” revolutionised our understanding of the Universe. In 1915, Einstein completed his general theory of relativity and in 1921 he was awarded the Nobel Prize in Physics for his explanation of the photoelectric effect.

Today, the practical applications of Einstein’s theories include the development of modern electronics, GPS systems, lasers, and medical imaging techniques. Recognized as TIME magazine’s “Person of the Century” in 1999, Einstein coupled his intellect with strong passion for social justice and dedication to pacifism, and gave the world both knowledge and pioneering moral leadership.

Einstein’s special theory of relativity

The special theory of relativity changed forever ideas about space and time, energy and mass. The theory reveals to us that one person’s interval of space is not the same as another person’s, and time runs at different rates for different observers travelling at different speeds. A moving clock appears to run slower and a moving object measures shorter in its direction of motion. It also tells us that the momentum of a moving object increases exponentially as its velocity increases until, at the speed of light, it becomes infinite. The theory leads to the idea that energy and mass, once thought to be two distinctly different properties, are equivalent and interchangeable.

The reason that these predictions are not obvious in everyday observations is that the effect at everyday speeds for average size objects that we tend to deal with is very small. The effects only really become apparent at speeds approaching that of light itself.

The postulates of Einstein’s special relativity

Einstein’s theory is based on two simple postulates:

1. The laws of physics are the same in all inertial frames of reference.
2. The speed of light in vacuum has the same value in all inertial frames of reference.

The first postulate implies that there is equivalence between all inertial frames. An inertial frame is a place in space that is not experiencing any accelerating forces. It includes objects at rest and those moving at constant velocity – but then we need to ask, what does ‘at rest’ mean? There is no way of knowing this. There is no experiment that we can perform to determine our velocity or to ascertain whether we are at rest. All we can do is measure our velocity relative to some other object or reference frame. As we watch a train rush by we could say it has a velocity of 100 km h^{-1} . But an observer on a similar train passing in the opposite direction who suddenly looks out the window will see the first train pass at 200 km h^{-1} . An observer in an aircraft travelling overhead will observe a completely different velocity, as would someone on the surface of the moon peering down through a powerful telescope. The Earth moves across the solar system at high speed but we have no way of knowing this as we sit at our desk where we imagine we are ‘at rest’.

Notes

Special relativity explained

Notes

There is also no such thing as an absolute reference frame. A fixed point in space in one frame is a moving point for another frame. People in two frames moving relative to each other will not agree about who is moving and at what velocity.

The first postulate was understood before Einstein. Einstein's revolutionary contribution is the new concept of the universality and constancy of the speed of light. The second postulate is the one that forces us to change the laws of physics as they were known before Einstein's relativity.

The universal speed limit

Surprising properties of light were discovered in the late nineteenth century. These properties form the basis of the theory of special relativity.

Experiments performed with radio and light waves emitted by pulsars, with light emitted from particles in accelerators, or with the light of gamma-ray bursts all show that the speed of the electromagnetic radiation does not depend on the frequency of the radiation, or on its polarisation, or on its intensity. After starting together and travelling together for thousands of millions of years across the Universe, light beams with different properties still arrive side by side. Other experiments show that the speed of light is the same in all directions of space.

But this invariance of the speed of light is puzzling. We all know that in order to throw a ball as fast as possible, we run as we throw it. We know instinctively that the ball's speed with respect to the ground is higher than if we do not run. We also know that hitting a tennis ball more forcefully makes it travel faster. But light behaves differently.

Experiments show that light emitted from a moving lamp has the same speed as light emitted from a resting one. One way to prove this is to look at the sky. The sky shows many examples of binary stars, stars that rotate around each other along ellipses. In some of these systems, we see the ellipses (almost) edge-on so that each star periodically moves towards and away from us. If the speed of light would vary with the speed of the source, we would see bizarre effects, because the light emitted from some positions would catch up with the light emitted from other positions. We would not be able to observe the elliptical shape of the orbits. However, such strange effects are not seen and perfect ellipses are observed. In other words, light in a vacuum will never travel faster than the speed of light. Experiments and theory show that no object can reach the speed of light. The speed of light is the universal speed limit; it is the maximum speed in nature. The velocity v of any physical system in nature is bound by c . The existence of an invariant limit speed c is not as surprising as we might think but, nevertheless it leads to many interesting results: it leads to observer-varying time and length intervals, to an intimate relation between mass and energy, to the existence of event horizons and to the existence of antimatter.

The complete theory of special relativity is contained within these statements.

- All light beams have the same speed.
- The speed of light in vacuum is invariant (it doesn't change).
- The speed of light is a universal limit speed.

Space and time

Because of Einstein's theory of relativity we can now say that space and time are related to each other by a common factor, the speed of light. Because it has the same value in all frames of reference it does not matter whether you are stationary with reference to a light source, travelling toward it or travelling away – the speed at which light reaches you is always the same. The constant velocity of light is used in our modern definitions of both time and distance.

Light moves extremely rapidly but with a finite speed. Today the speed of light is specified with a precision of nine significant figures. The metre is now defined in terms of the speed of light. Since 1983 the metre has been defined by international agreement as the distance travelled by light in vacuum during a time interval of $1/299,792,458$ of a second. This makes the speed of light exactly $299,792.458 \text{ km s}^{-1}$.

Time is also defined in terms of the properties of light. One second is defined as the duration of $9\,192\,631\,770$ periods of the wavelength of light emitted by caesium-133 produced by electron transitions between the two hyperfine levels of the ground state of that atom.

Some astronomical distances are defined by their relationship to the speed of light. An example is the light year – the distance traversed by a photon in one year of travel through vacuum.

The light that we see from our nearest neighbour star, Proxima Centauri, was transmitted 4.2 years ago so Proxima Centauri is 4.2 light years from Earth. Light produced by the Sun takes approximately 8 minutes to reach the Earth, so the diameter of the Earth's orbit around the Sun is about 16 light minutes.

Space-time

Space-time (also known as the space-time continuum) is a model that combines space and time into a single model. The Universe is usually interpreted from a Euclidean space perspective where space consists of three dimensions often described on a Cartesian system with x, y and z axes. In space-time a “fourth dimension” is introduced to enable time information to be included. Any event can now be described by knowing its location in space and the time at which it occurred. By multiplying time by the speed of light the time dimension will have the same units as the space dimensions.

Four dimensional space is difficult to visualise and even more difficult to draw on a two-dimensional screen or piece of paper. To simplify we can show just a single space dimension, such as distance in one direction which we could call ‘x’ and a time axis in the same units as the ‘x’ axis. For example if ‘x’ is measured in metres and time in seconds then time can be multiplied by the speed of light ‘c’ to give units of metres.

Figure 14.2 shows a space-time plot for a car travelling at 20 m s^{-1} (about 72 km h^{-1}). Notice that the horizontal axis is severely squashed. If the two axes were scaled equally the horizontal axis would be about 1500 km long.

For particles travelling at speeds close to the speed of light we can produce an equally scaled plot as shown in Figure 14.3. This graph shows the space-time trajectories of two particles – one travelling at $0.95c$ and another at $0.99c$. The solid line represents the speed of light so all events that we can experience will occur in the space-time region below this line. The space-time plot for the car in the previous figure is shown also – its space-time line is virtually horizontal.

The factor v/c is often referred to as β where: $\beta = \frac{v}{c}$

Special relativity explained

Notes

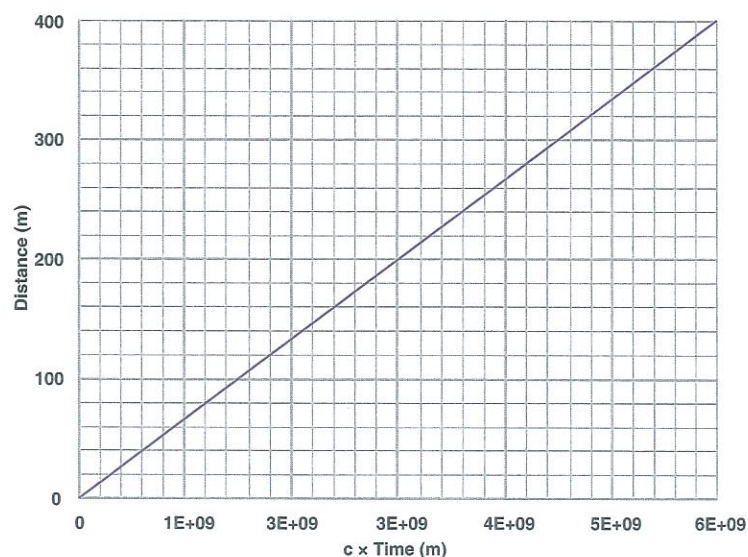


Figure 14.2: A simple space-time plot for a car travelling at 20 m/s (about 72 km/hr).

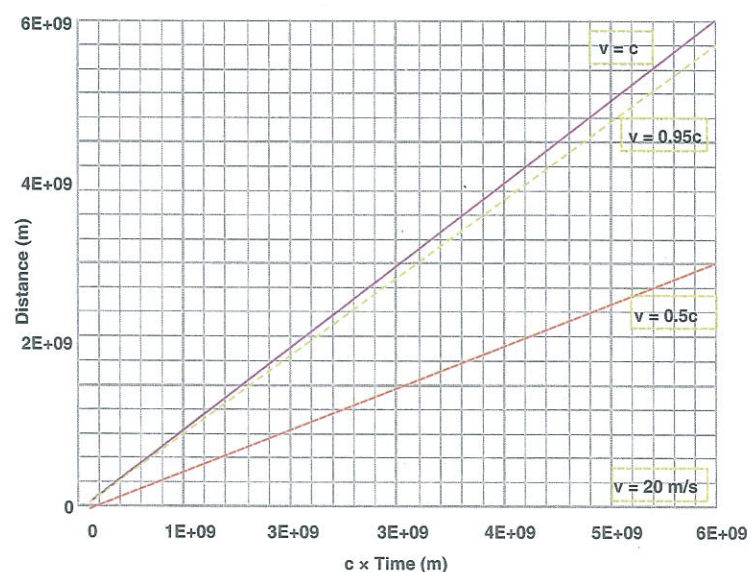


Figure 14.3: The space-time trajectories of two particles – one travelling at $0.95c$ and another at $0.99c$. The solid line represents the speed of light.

Hermann Minkowski

A mathematically rigorous description of space time was developed by mathematician Hermann Minkowski in the early 20th century. Now known as 'Minkowski space' or 'Minkowski space-time' this model comprises a four-dimensional manifold where the space-time interval between any two events is independent of the inertial frame of reference in which they are recorded. Although initially developed for Maxwell's equations of electromagnetism, the mathematical structure of Minkowski space-time was shown to be useful when adapted to Einstein's theory of special relativity, and is the most common mathematical structure on which special relativity is formulated. Because it treats time differently Minkowski space differs from simple four-dimensional Euclidean space. The model has significantly simplified a large number of physical theories.

Mass and energy

Mass can be defined as the measure of how much matter an object or body contains - the total number of sub-atomic particles (electrons, protons and neutrons) in the object. But also mass can be defined by Newton's second law, $F = ma$. It can be seen as the resistance to acceleration specifying the amount of force required to cause a body to accelerate.

Energy is the measure of a system's ability to make changes or in mechanics, the ability to perform "work". It exists in many forms: potential, kinetic, thermal etc. The law of conservation of energy tells us that energy can neither be created nor destroyed; it can only be converted from one form to another. It is the total amount of energy that is conserved. If you drop a rock from a bridge the rock has kinetic energy the moment it starts to move. Just before you dropped the ball, it had only potential energy. As the rock moves, the potential energy is converted into kinetic energy. Likewise, when the rock hits the ground, some of its energy is converted to thermal energy. If you find the total energy for the system, you will find that the amount of energy for the system is the same at all times.

The unification of energy and mass

The concept of mass has been fundamental to physics. Its definition goes back to Galileo and Newton who said that mass was that property of a body that enables it to resist externally imposed changes to its motion. Newton used mass to define momentum and force: he defined a body's momentum as $p = mv$ (where v is its velocity), and he defined force in terms of an object's acceleration: $F = ma$.

This definition of mass was applied in a straightforward way for almost two centuries. Then Einstein arrived on the scene and, in his theory of special relativity, he changed this way of thinking forever. Undoubtedly the most famous equation ever written is $E=mc^2$. This equation says that energy is equal to the mass of an object times the speed of light squared. This means that energy and mass are interchangeable – they are manifestations of the same thing. Since the speed of light is constant, an increase or decrease in the system's mass is proportional to an increase or decrease in the system's energy. The law of conservation of energy and the law of conservation of mass can be combined to give one law, the law of conservation of energy.

It is often stated that an object's mass increases as it approaches the speed of light. For example, consider a proton accelerated in a particle accelerator. The following occurs:

- 1) Energy must be added to the system to increase its speed.
- 2) More of the added energy goes towards increasing the particle's resistance to acceleration.
- 3) Less of the added energy goes into increasing the particle's speed.
- 4) Eventually, the amount of added energy required to reach the speed of light would become infinite. The proton never reaches the speed of light no matter how much energy is added. The speed of light can be considered the universal speed limit.

In step 2, the system's resistance to acceleration is a measurement of an object's mass, which increases when the object speeds up. This is called relativistic mass. It should be noted that many physicists have stopped using this concept of mass and now consider this effect in terms of an increase in momentum and energy. The Newtonian definition of mass still holds for a body at rest, and so has come to be called the body's *rest mass*, often denoted m_0 .

Energy and Momentum

In classical mechanics, kinetic energy E_k and momentum p are expressed as

$$E_k = \frac{1}{2}mv^2$$

$$p = mv$$

Special relativity explained

Notes

These relationships work well at low speeds and for objects of average mass, but at high speed we need to apply the relativistic corrections:

$$E_k = \gamma mc^2$$

$$p = \gamma mv$$

Where γ is the Lorentz factor as described on page 180.

In some relativity textbooks, the so called “relativistic mass” $m = \gamma m_0$ is used as well. Many authors prefer to use the expressions of relativistic energy and momentum to express the velocity dependence in relativity, which provide the same experimental predictions.

Relativistic energy and momentum increase significantly at speeds approaching the speed of light and at these speeds enormous amounts of energy are required. Therefore no massive particle can ever reach the speed of light.

Photons

We have seen that as we accelerate a massive particle up to the speed of light we find that the energy increases rapidly with little change in the particle's velocity. Einstein's formula predicts that, at the speed of light, the energy of the particle would become infinitely great. In other words, it would require an infinitely large amount of energy to accelerate the particle to the speed of light. This is, quite obviously, impossible so how then does a photon manage to travel with the speed of light? The answer is that photons have zero ‘rest’ mass, which means that the energy of a photon is all kinetic energy. If a photon is forced to come to rest in some absorbent material, it simply ceases to exist.

If individual photons have a definite energy and are able to behave like particles, we should expect them to carry momentum. It is natural to think that light carries energy as the Earth receives tremendous amounts of solar energy from the Sun. Einstein had used the hypothesis that light could be regarded as a stream of individual packets of energy to explain the ‘photo-electric effect’, the fact that light shining on a metal surface can cause electrons to be ejected. Einstein showed that, if Planck's law of black-body radiation is accepted, photons must also carry momentum $p = h/\lambda$. This photon momentum was observed experimentally by Arthur Compton, for which he received the Nobel Prize in 1927.

To find a photon's momentum we can combine Planck's formula ($E = hf$) and Einstein's mass – energy formula ($E = mc^2$):

$$p = mc = \frac{E}{c} = \frac{hf}{c} = \frac{h}{\lambda}$$

Simultaneity

How do we know two events happen at the same time? When we see a flash of lightning we know its associated sound, thunder, will arrive shortly after. This time delay is a result of the different rates at which the information is conveyed to us – the flash of lightning travels at the speed of light, the thunder at a pedestrian pace equal to the speed of sound. The time delay can be used to calculate how far away a lightning strike is from the place where we are located.

If we observe two events at locations where we know the distance to our point of observation we can carry out a simple calculation to determine if the two events are simultaneous – that is, did they happen at exactly the same time? But this is not so easy in Einstein's world. Whether or not two events occur simultaneously depends upon your frame of reference. The time order of events that are close together in time but distant in space can be different in different frames.

It was common for Einstein to create simple thought experiments to help illustrate the concepts of his theories. One such thought experiment illustrates the point of simultaneity. Examples similar to this are discussed in many textbooks.

Imagine a train moving with uniform high velocity. A light source located in the centre of the train transmits a light pulse in all directions. A detector (or clock) at the front and rear of the train record the moment a pulse of light is detected. Marie, an observer on board the train recognises that the pulses strike each detector simultaneously. At this moment an observer named Albert on a 'stationary' platform observes the train pass by just as the light pulses are emitted.

As light travels out from the source, according to Albert, the stationary observer, it must be travelling at the speed of light in both directions. After the pulses are emitted the rear of the train has moved closer to the source and therefore has less distance to travel. The forward going pulse has further to travel. Albert perceives that the light pulse has reached the rear detector first and he concludes that the two pulse detection events are not simultaneous.

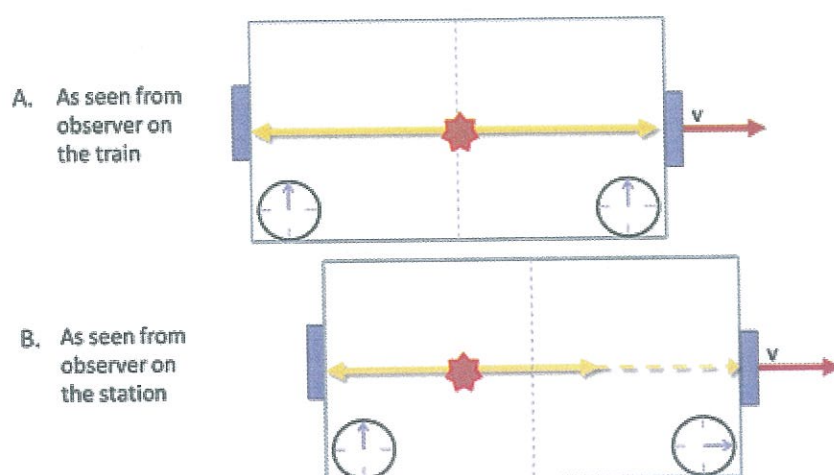


Figure 14.4: A light source located in the centre of the train transmits a light pulse in all directions. An observer on the train sees the pulses strike each detector simultaneously. A stationary observer sees the light pulse reach the rear detector first.

Special relativity explained

Notes

This thought experiment demonstrates that the two events that appear to be simultaneous to one observer do not appear to be simultaneous to another observer. In other words, two events that are simultaneous in one reference frame in general do not appear to be simultaneous in a second frame moving relative to the first. Simultaneity is not an absolute concept but rather one that depends on the state of motion of the observer.

Einstein's thought experiment demonstrates that two observers can disagree on the simultaneity of two events. Visualising events in space-time can help us understand this concept. Imagine you are looking at a row of trees in an orchard. Standing at a position at the end of the row you can see the trees in a line in front of you and you could say they all share the same y -coordinate. From another vantage point, say some distance perpendicular to the line, we can see the trees separated by 2 metres. But if you think of time as just another "direction" you can believe that your time coordinate could have one value in one reference frame, and another value in a different frame.

Time dilation and length contraction follow from this in a straightforward way. And they are much easier to visualise if you can see that time is another "direction", and therefore is relative to the observer, just like position is.

Discussion: Sound waves

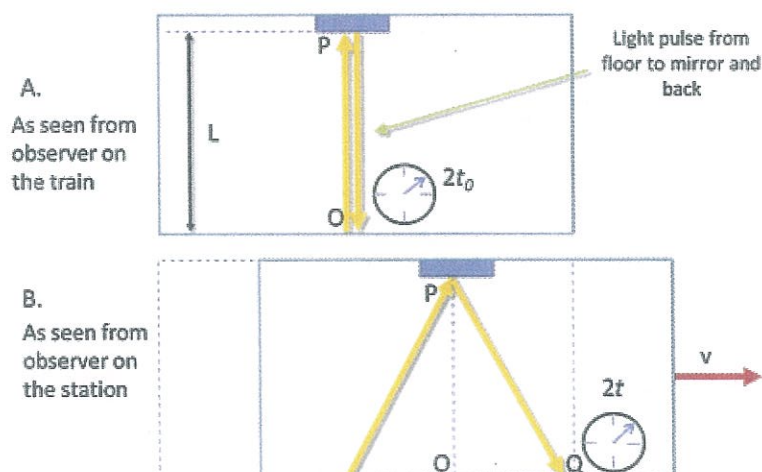
Consider the previous experiment but with the light source replaced by a sound source. When the sound reaches the two detectors a flash of light is given off. Will the two observers agree on the simultaneity of the sound pulse? Discuss.

Time dilation

Another typical thought-experiment is the analysis of the "light-clock". The basic idea of a light-clock is to use the distance travelled by a pulse of light and the known speed of light to mark out intervals of time. The following example is discussed in many textbooks and can be used to derive the time dilation formula. The thought experiment proceeds as follows:

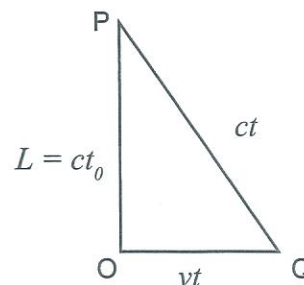
Albert, inside a train carriage, sets up an experiment where he directs a pulse of light to a mirror on the roof of the train. The beam returns to its starting position and the time for the return trip is measured. This is shown in part 'A' of the following diagram.

The next step in this experiment is to imagine the train moving at high velocity through the station from left to right and consider how the light pulse would appear to Marie who is standing on the platform of a railway station. She would observe the path of the light pulse as two diagonal paths as shown in part 'B'.



The triangular path OPQ is shown below. Knowing that for constant velocity - distance = velocity multiplied by time - we can determine the length of the sides of the triangle:

From part A the length L is equal to the velocity of light c multiplied by the time taken, t_0 : That is, $L = ct_0$. Similarly the diagonal edge of the triangle is ct and the base of the triangle is vt as shown. Note that t_0 is known as the 'proper time' and is the time measure in the frame containing the light clock, in this case the train carriage and t is the time measured relative to the proper time. The length of the light path PQ as viewed by Marie on the platform is longer than the path PO as observed by Albert in the train carriage. The speed of light ' c ' is the same for both observers so therefore time as measured by Albert, the moving observer, must be ticking 'slower' than time as observed for Marie, the stationary observer.



Notes

If we analyse the triangle using Pythagoras' theorem we get:

$$(ct)^2 = (vt)^2 + (ct_0)^2$$

$$(ct_0)^2 = (ct)^2 - (vt)^2$$

$$t_0 = t \sqrt{1 - \frac{v^2}{c^2}}$$

$$t = t_0 \frac{1}{\sqrt{1 - \beta^2}}$$

Using $\gamma = \frac{1}{\sqrt{1 - \beta^2}}$ we get: $t = \gamma \cdot t_0$

Here t_0 is the time as measured in the frame of the moving 'light-clock' and t is the time as measured by the stationary observer. γ is the Lorentz factor and will be described in the next section.

The Lorentz Transform

Hendrik Lorentz (18 July 1853 – 4 February 1928) was a Dutch physicist who shared the 1902 Nobel Prize in Physics with Pieter Zeeman for the discovery and theoretical explanation of the Zeeman Effect - the splitting of a spectral line into several components in the presence of a static magnetic field. He also derived the transformation equations subsequently used by Albert Einstein to describe space and time. The Lorentz transform adapted from his work is a way of comparing observations from different reference frames. It is a way to bring observers of different velocities at different places together so they can 'compare notes'. It takes into account the fact that the speed of light is constant, and finite, but distance and time are not constant.

For example, if you are observing a building from the front entrance and a friend two streets away observes the same building how do you compare your observations about the size and shape of the building? You need to make an adjustment, some sort of a transform, to reflect your different viewpoints. In special relativity this becomes complicated since neither distance nor time are constant between observers moving relative to each other.

In the Lorentz transform measurements in the direction of motion are adjusted by a factor γ (gamma) known as the Lorentz factor. Its value is given by:

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} = \frac{1}{\sqrt{1 - \beta^2}}$$

Where v is the relative velocity of the two reference frames and c is the speed of light. Gamma is always greater than one because v is always less than c .

Special relativity explained

Notes

The following graph is a plot of gamma and $1/\gamma$ for speeds up to the speed of light.

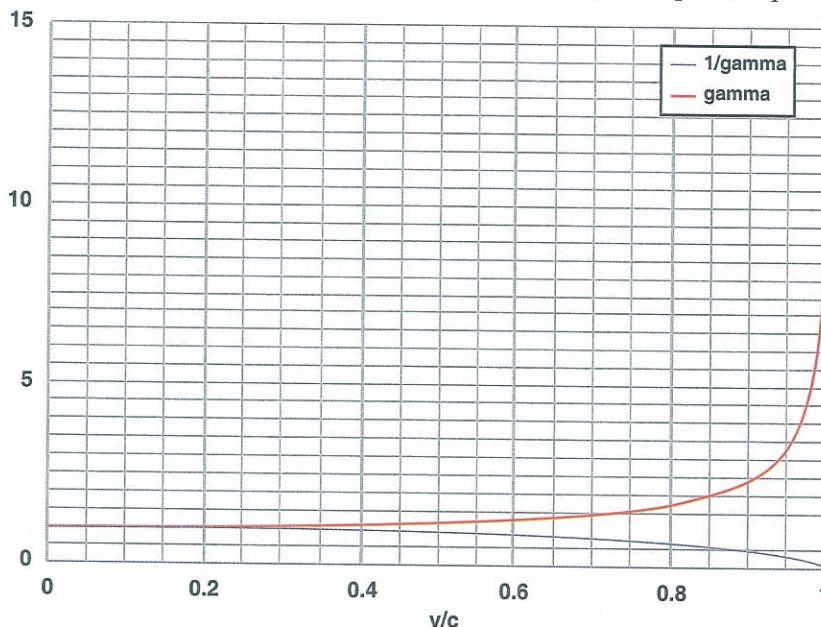


Figure 14.6: A plot of gamma and $1/\gamma$ for speeds up to the speed of light

The relativity of length

As objects move through space-time, space as well as time changes. As an object travels at relativistic speeds it contracts or gets shorter. The term “length contraction” appears in many textbooks but what does it mean? Do fast-moving objects really shrink? Or does an object only appear to shrink, or does the observer only “perceive” a contraction?

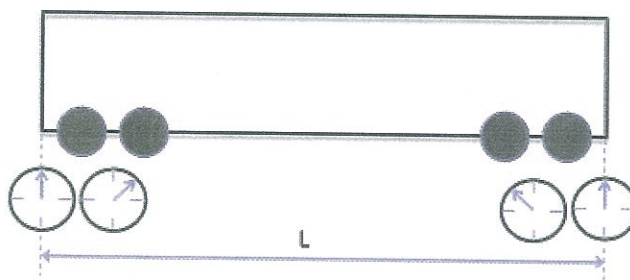


Figure 14.7 Measuring the length of a moving train by locating its front and back at the same time

Figure 14.7 shows the difficulty of trying to measure the length of a moving train by locating its front and back at the same time. Because simultaneity is relative and it enters into length measurements, length should also be a relative quantity. It is.

A suitable thought experiment is to compare the length of the train with a known external length such as the length of a tunnel.

Marie is on the train, with synchronised clocks placed at each end. Albert is on the station. By carefully setting up the clocks Marie sees the front clock strike noon as the train exits the tunnel and the back clock strike noon as the train enters the tunnel. To Marie the clocks strike noon simultaneously and she concludes that the train fits exactly in tunnel. For Albert the back clock strikes noon first and the front clock a little later so he concludes that the train is shorter than the tunnel.

Th
ob
de
po
tw
rej
kn

M
of

If
by
the
 $\frac{L}{L_0}$
 L
Th
Do
the
me
is t
do
obs

Th
No
0.9
mu
res

To
oth
obj
a re
the

Fig
has